

Indian Karting Race 2025

Season 10.0



Team Name: **Marlin Racing**

Team ID: **IKR25-EV-2735**

Report Title: **MARK 1 Design Report for The
Indian Karting Race 2025**

Report Subtitle: **Design of an Electric Race Kart**

ABSTRACT: This report documents the complete design, analysis, and validation of an electric go-kart developed for the Indian Karting Race (IKR) 2025. It comprehensively covers all critical subsystems including the chassis, steering, transmission, braking, motor, motor controller, battery, and electrical wiring each developed in compliance with the technical and safety standards outlined in the IKR 2025 rulebook. The chassis employs AISI 4130 Chromoly steel, optimized using CAD modeling and Finite Element Analysis (FEA) for maximum strength and safety. A mechanical four-bar linkage steering system was designed for reliability and minimal play, incorporating Ackermann geometry for improved handling. The transmission utilizes a chain-sprocket drive with strength-calibrated shafts, while a hydraulic disc braking system ensures safe deceleration with validated stopping distances. The powertrain features a 4 kW BLDC motor and a programmable sinusoidal controller integrated with safety interlocks. A 60V, 90Ah Li-ion battery pack, configured with BMS and short-circuit protection, powers the system. All components were selected to optimize performance, efficiency, and manufacturability, resulting in a competition-ready kart aligned with IKR 2025 performance and safety benchmarks.

CHASSIS DESIGN REPORT:

● Introduction

The chassis is the structural backbone of a racing kart, responsible not only for supporting all mechanical components but also for ensuring driver safety, performance, and compliance with competition standards. In the context of the Indian Karting Race (IKR) 2024, chassis design assumes utmost importance due to the technical and safety regulations laid out in the official rulebook. This report presents a comprehensive overview of the design, analysis, and validation of a high-performance go-kart chassis developed specifically for IKR 2025.

The design process was guided by a balance of critical engineering objectives which were structural strength, lightweight construction, cost-effectiveness, and regulatory compliance. Key aspects such as material selection, geometric configuration, safety features, and mechanical performance were carefully considered to develop a chassis that meets the demands of competitive kart racing. The design adheres to rulebook constraints. The chassis is the structural backbone of a racing kart, responsible not only for supporting all mechanical components but also for ensuring driver safety, performance, and compliance with competition standards, governing materials, tubing dimensions, roll hoop specifications, bumper and jack point placement, minimum ground clearance, and driver compartment safety.

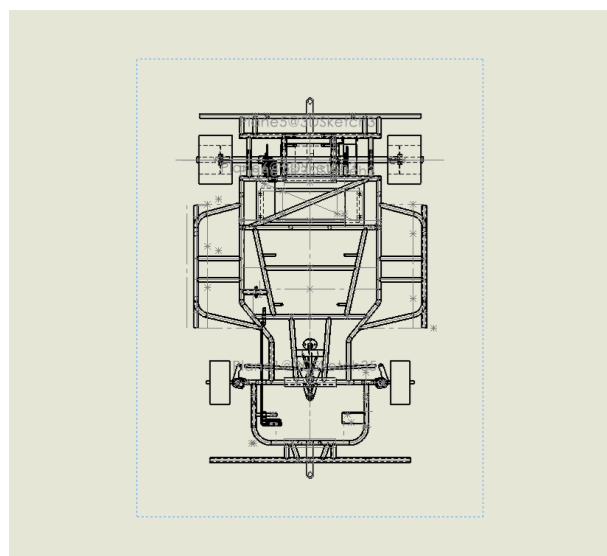
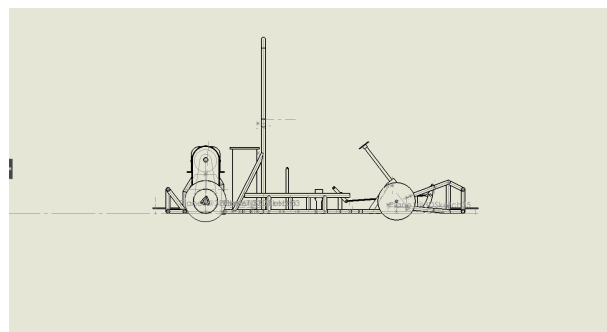
Advanced CAD modeling and Finite Element Analysis (FEA) techniques were employed to simulate real-world loading scenarios, including impact and dynamic forces. These analyses helped validate the structural integrity and crashworthiness of the chassis design. AISI 4130 Chromoly steel was selected as the primary construction material due to

its superior strength-to-weight ratio and proven performance in motorsports applications.

This report outlines the constraints, methodology, design parameters, material choices, and safety validation that went into the development of the chassis. Through an iterative, engineering-driven approach, the final design achieves a robust, competition-ready structure that satisfies all functional, ergonomic, and safety requirements outlined by IKR 2024. Its governing materials, tubing dimensions, roll hoop specifications, bumper and jack point placement, minimum ground clearance, and driver compartment safety.

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- **Rulebook Constraints**

The Indian Karting Race (IKR) 2024 rulebook outlines a comprehensive set of chassis design rules to ensure safety, structural integrity, and fairness in competition. The chassis must be constructed using seamless circular steel or aluminium tubes with an outer diameter between 1 to 2 inches and a minimum wall thickness of 1.2 mm. Use of materials like leaf springs, I-bars, or solid bars is strictly prohibited. A 3mm drill hole is required in the main roll hoop to inspect wall thickness. The chassis must include a capped open-end tube and comply with design mandates including placement of bumpers, seat compartments, engine housing, and specified clearances between components.

Additional chassis-related design requirements include mandatory roll hoops fabricated from a single bent tube, extending at least 3 inches above the tallest driver's helmet. The roll hoop must be braced with straight steel tubes, and no wrinkles or bends are allowed in the structure. Bumpers must be mounted on all sides using single seamless tubes and have at least four chassis attachment points, with foam insulation if not covered by bodywork. The kart must also include clearly painted jack points (yellow) at the front and rear, each constructed from flat steel plates of specified dimensions, and hitch points (orange) for push rod attachment, positioned on the chassis (not bumpers) with a 25.4 mm diameter hole.

The driver's compartment must prioritize safety and ergonomics. The seat must be securely mounted with four bolted points, positioned 2 inches away from the firewall. A robust foot guard is required to prevent foot injuries, and firewalls made from at least 1.5 mm thick aluminum must completely isolate the driver from the engine or battery compartment. These must be rigid, sealed, and free from holes or cut-outs. Furthermore, the minimum ground clearance is 1 inch and maximum 5 inches (measured with driver), and maximum turning radius is limited to 3 meters. Lastly, specific guidelines for wheelbase (minimum 40 inches) and track width (minimum 65% of wheelbase) must be followed to ensure stability and control.

- **Methodology**

The chassis design methodology for the Indian Karting Race (IKR) 2024 was developed based on a structured engineering approach in accordance with the technical specifications outlined in the official rulebook. The process began with a thorough analysis of all design constraints and safety requirements, focusing on

material selection, structural layout, and integration of essential components. The primary chassis material was selected as seamless circular steel tubing, adhering to the rule-mandated outer diameter range of 1 to 2 inches and a minimum wall thickness of 1.2 mm. The chosen material ensured sufficient strength, manufacturability, and resistance to dynamic loads encountered during track events.

Initial conceptual designs were drafted using CAD software to evaluate possible frame geometries that offered optimum strength-to-weight ratio and packaging efficiency. The roll hoop was designed using a single bent tube of the same chassis material and section, welded directly to the frame, and extending a minimum of 3 inches above the helmet of the tallest driver as per C.2.6.7. The design also incorporated two roll hoop braces oriented rearward, constructed from straight closed-section tubing. Additional structural elements such as front and rear jack points (each 300 mm x 50.8 mm x 5 mm), hitch points with a 25.4 mm hole diameter, and front/rear/side bumpers were designed in compliance with sections C.2.4, C.2.5, and C.2.7 respectively.

A minimum ground clearance of 1 inch and maximum of 5 inches (with driver seated) was maintained, and the overall wheelbase was designed to exceed the minimum requirement of 40 inches, ensuring vehicle stability and control. The track width was maintained at a value greater than 65% of the wheelbase as per C.2.10. To verify structural integrity, Finite Element Analysis (FEA) was conducted on the chassis model to assess stress distribution, deformation, and potential failure points under simulated dynamic loads such as braking and cornering. All critical zones including the driver's seat mounts, steering column support, and engine housing were analyzed for stiffness and safety. The final chassis design met all dimensional, material, and safety requirements, providing a structurally sound platform for the rest of the vehicle systems.

- **Material Selection**

For the chassis of our go-kart, we selected AISI 4130 Chromoly steel due to its excellent balance of strength, weight, and manufacturability. AISI 4130 is a low alloy steel containing chromium and molybdenum as strengthening agents, which provides significantly higher tensile strength and fatigue resistance compared to standard mild steels. This enhanced mechanical performance is critical in a racing application where the chassis must withstand dynamic loads such as braking, cornering, and impacts while maintaining structural integrity. Its yield strength typically exceeds 460 MPa, offering a high safety margin without requiring excessive material thickness, which helps keep overall weight within optimal limits.

Another key factor in choosing AISI 4130 was its proven weldability and availability in seamless round tubes, as required by the IKR 2024 rulebook. This material is commonly used in motorsports and aerospace structures, where reliability and precision are crucial. It allows for clean, strong welds without the need for pre-heating when working with thinner sections, which is advantageous during fabrication. Furthermore, the density and strength-to-weight ratio of 4130 make it ideal for maintaining chassis stiffness while reducing unnecessary mass, improving acceleration and handling characteristics. Overall, AISI 4130 offered the ideal combination of strength, safety, and workability for our chassis design.

• Design Parameters

The chassis of our racing kart has been meticulously designed to ensure high performance, durability, and responsive handling characteristics on the track. The kart features a wheelbase of 1300 mm, which strikes a calculated balance between straight-line stability and tight cornering capability. A front track width of 1050 mm and a slightly wider rear track width of 1100 mm contribute to enhanced lateral stability and improved traction during aggressive maneuvers. These dimensions help optimize weight transfer and minimize oversteer or understeer tendencies. The overall dimensions—2033 mm in length and 1290 mm in width—were chosen to remain within competition guidelines while allowing adequate space for component packaging and driver ergonomics.

Static ground clearance is set at 2.23 inches (approximately 56.6 mm), chosen to lower the center of gravity without risking contact with the track surface during dynamic motion. This low profile aids in better handling and increased downforce efficiency at speed. The chassis frame is constructed using 1-inch ISO-standard hollow steel tubing, selected for its high strength-to-weight ratio, ease of fabrication, and compliance with safety regulations. The wheel and tire configuration includes 240 mm diameter wheels at the front and 250 mm at the rear, providing a slight stagger that supports steering responsiveness and rear-end grip. This setup contributes to a stable yet agile platform suited for competitive kart racing.

• Design Considerations

The chassis design of the racing kart was centered around achieving an optimal balance between performance, safety, and structural efficiency. One of the most critical parameters considered was the wheelbase, set at 1300 mm. This dimension was chosen to provide a stable platform during high-speed operation while maintaining responsive handling in corners. A

well-balanced wheelbase helps control weight transfer during acceleration and braking, thereby improving overall vehicle dynamics and driver control.

Equally important were the track widths, with the front track at 1050 mm and the rear track slightly wider at 1100 mm. This deliberate difference aids in achieving a neutral to slightly understeering handling characteristic, which is preferred in competitive karting for better predictability and control. The wider rear track also enhances lateral grip and reduces the likelihood of rear-end instability during aggressive cornering. The static ground clearance, set at 2 inches, was carefully selected to lower the center of gravity without risking contact with the ground during operation. This contributes significantly to minimizing body roll and improving chassis responsiveness.

The choice of 1-inch ISO-standard hollow steel tubing for the chassis frame was made to ensure structural integrity while keeping weight to a minimum. The tubing offers a high strength-to-weight ratio and is readily manufacturable, making it ideal for a racing environment. Additionally, the wheel diameters—240 mm at the front and 250 mm at the rear—were selected to improve forward traction and steering response, supporting consistent performance across varied driving conditions. These dimensions, together, form the backbone of a design that meets the dual objectives of competitive performance and safety compliance.

• Roll Cage Analysis

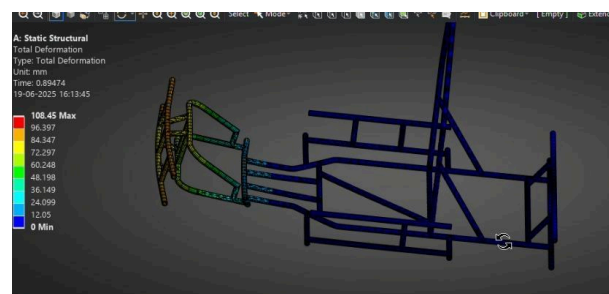


Figure 1.1 Frontal Impact Test Results

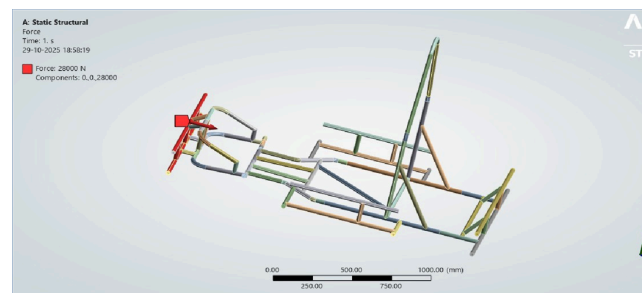


Figure 1.2 Frontal Impact Test (Force Applied)

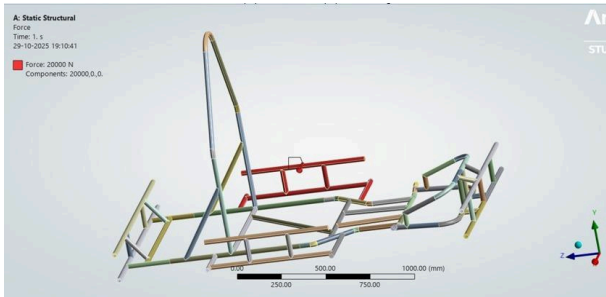


Fig 1.3 Side impact test (force applied)

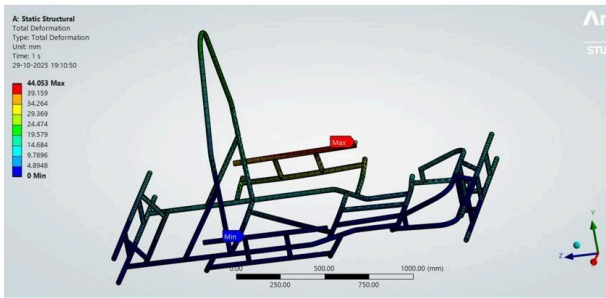


Fig 1.3 Side Impact test result

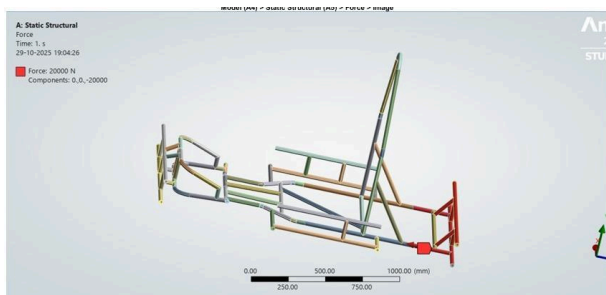


Fig 1.4 Rear impact test (force applied)

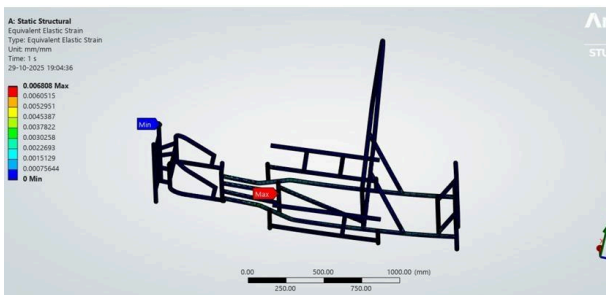


Fig 1.5 Rear impact test result

- The maximum load applied in the simulation was **28,788 N**, derived from a frontal crash at **70 km/h (19.44 m/s)**. This was treated as a **static structural load** to simulate worst-case crash conditions. Under this load, the maximum **von Mises stress observed in the chassis was 281 MPa**, with a corresponding **minimum factor of safety (FOS) of 1.43** in the most critical regions. These values fall within acceptable safety margins for the material used—**AISI 4130 steel**, which has a yield strength of approximately **413 MPa**.

In our kart's design, the chassis skeleton itself functions as an integral part of the roll cage, combining both structural rigidity and occupant protection in a single frame. This design choice simplifies manufacturing and increases torsional stiffness. The chassis was modeled in SolidWorks and imported into ANSYS, where AISI 4130 properties were applied. The total mass of the chassis-roll cage assembly was calculated to be approximately **45.8 kg**. Static structural analyses were conducted for **frontal, rear, and side impact conditions**, applying directional loads to simulate potential crash scenarios based on the kart's top speed. All simulations used bonded contacts and fixed supports at wheel mounting points to replicate realistic boundary conditions.

- To calculate the theoretical crash load, we used the kinetic energy method:

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2} \times 45.816 \times (19.44)^2 \approx 8,636 \text{ J}$$

Assuming a conservative **crumple zone deformation of 0.3 meters**, the impact force was estimated as:

$$F = \frac{E_k}{d} = \frac{8,636}{0.3} \approx 28,788 \text{ N}$$

- This force was directionally applied in the simulations to assess the chassis response under real-world crash conditions.
- Under frontal impact, the chassis structure showed higher stress concentrations around the front cross members and the steering column base, with a peak stress of **281 MPa**. Rear and side impacts yielded slightly lower stress values, with minimum FOS above **1.6**. The areas of maximum deformation were concentrated in non-critical zones, meaning no permanent deformation occurred in regions essential to occupant safety. The deformation results also confirmed that the roll cage design effectively channels forces away from the driver compartment.

- The simulation results validate the design by confirming that all critical components remain within the material's elastic limit under impact. With all FOS values exceeding **1.4**, the chassis-roll cage assembly satisfies safety expectations for competitive karting while minimizing unnecessary weight. The integration of theoretical analysis with FEA results provided a robust validation of our design methodology.

(NOTE: The maximum Number of pages that can be used for Chassis section is limited to 5 including this page. Complete report should be in the same IEEE format)

STEERING DESIGN REPORT:



Introduction:

The steering system allows the driver directional of the vehicle. In compliance with the rulebook, the steering system is purely mechanical, with no electronic or steer-by-wire components, and all parts must be made of metal, as per rulebook guidelines.

In this design, we have used a simple and reliable four-bar linkage steering mechanism. The system includes a steering wheel connected to a steering column, which is attached to a pitman arm. The pitman arm drives two tie rods, which are connected to the front wheel uprights. When the driver turns the steering wheel, the movement is transferred through the linkage to the front wheels, causing them to turn.

This type of system is lightweight, easy to manufacture, and provides the control needed during sharp turns and high-speed maneuvers. Special attention has been given to minimize play in the steering, maintain a turning radius below the 3-meter maximum limit, and ensure full compliance with the IKR 2025 rules.

Rulebook Constraints:

C.3.6.1 Steering Mechanism

C.3.6.1.1 Teams can use any type of mechanical steering mechanism. (Rack and Pinion/ Steer by wire is excluded).

C.3.6.1.2 The steering wheel must be mechanically connected to the wheels, i.e. “steer-by-wire” or electrically actuated steering is prohibited.

C.3.6.1.3 All parts of steering system (steering column, track rod, tie rod, etc.) must be of metal except Steering wheel.

C.3.6.1.4 Carbon fiber, nylon, plastic etc. are not allowed.

C.3.6.2 Steering Wheel

C.3.6.2.1 The steering wheel must have a continuous perimeter that is near circular or near oval, i.e. the

outer perimeter profile can have some straight sections, but no concave sections. “H” or cut-out wheels are not allowed.

C.3.6.2.2 Allowable steering system free play is limited to Seven degrees (7°) total measured at the Steering wheel.

C.3.6.3 Minimum dimension of steering wheel

C.3.6.3.1 Full circle: outer diameter (O.D) of 10 inches

C.3.6.3.2 Elliptical: Major axis diameter of 10 inches and minor axis diameter of 8 inches.

C.3.6.4 Steering Lock

C.3.6.4.1 The steering system must have positive steering stops that prevent the steering linkages from locking up (the inversion of a four-bar linkage at one of the pivots).

C.3.6.4.2 The stops may be placed on the uprights and must prevent the tires from contacting body, or frame members during the track events.

Dimension Parameters:

Front Track Width: 1050 mm

Wheel base: 1299 mm

Distance between uprights: 850 mm

King Pin Inclination : 8 degrees

Negative Caster: 8 degrees

Steering Geometry:

A four-bar linkage with a Pitman arm in a go-kart steering system offers a reliable, mechanical way to convert steering input into wheel movement, completely meeting the rulebook's requirement for an all-metal, non-electronic system. The kart is effectively steered by the steering wheel rotating the steering column, which in turn activates the Pitman arm. This arm pivots and moves the drag link (center link), which in turn pushes or pulls the tie rods attached to the wheel spindles. The fixed frame, Pitman arm (input) and tie rods (output links) all function in tandem with one another in this traditional four-bar mechanism design. As mandated by rule C.3.6.4.1, it is essential to have appropriate steering stops on the uprights or the Pitman arm itself to avoid over-rotation or linkage inversion. Since materials like plastic, carbon fiber, or nylon are prohibited under C.3.6.1.3 and C.3.6.1.4, all parts, including the steering column, Pitman arm, drag link, and tie rods, must be made of metal. As long as the steering wheel measurements and the 7° free play also adhere to the rules outlined in C.3.6.2 and C.3.6.3, this setup is straightforward and ideal for the high responsiveness needed for the event.

Design Considerations:

The steering system has been designed to balance performance, safety, simplicity, and compliance with the IKR 2025 rulebook. The following considerations were taken into account during the design phase:

1. Compliance with IKR 2025 Rules

Only mechanical steering mechanisms are allowed. Steer-by-wire or electronic systems are strictly prohibited.

All structural parts (steering column, tie rods, uprights) must be metallic. Materials like nylon, plastic, or carbon fiber are not permitted.

Maximum steering free play allowed is 7°, which guided the tolerances in bearing and joint selections. The steering wheel must be circular or elliptical with minimum sizes (10" diameter for full circle or 10"x8" for elliptical), which dictated the wheel selection.

2. Type of Mechanism

A four-bar linkage was selected for its simplicity, compactness, and proven performance in kart applications.

The mechanism ensures reliable and consistent steering response, with manageable fabrication and easy servicing.

3. Steering Geometry



Ackermann Geometry was incorporated to reduce tire scrub during cornering. This improves handling and tire wear.

The positions of the steering arms and tie rod mounting points were optimized in CAD to closely match the ideal Ackermann curve.

4. Turning Radius

The entire system was designed to ensure a turning radius of less than 3 meters, which is the maximum allowed by the IKR rules.

Lock stops were added to the spindle arms to prevent oversteering and mechanical binding.

5. Steering Stops

Mechanical steering locks were included at the uprights to avoid the possibility of the linkage inverting—a critical safety requirement per IKR regulations.

6. Ergonomics and Packaging

Steering wheel placement was optimized for driver comfort, visibility, and natural arm positioning.

The steering shaft angle was chosen to provide a comfortable wrist angle while allowing adequate knee clearance.

The tie rods were routed carefully to avoid collision with suspension or chassis members during full travel.

7. Mounting and Serviceability

All mounts use metal brackets and lock nuts to prevent loosening due to vibration.

The steering column is supported by bushings/bearings in housings that can be easily accessed for maintenance or replacement.

8. Safety

Steering components are designed with a minimum factor of safety of 2.0, using mild steel or equivalent.

Sharp edges were removed or rounded off to ensure driver safety.

Component Selection:

Part Name	Quantity	Material
Steering Wheel	1	M.S.
Steering Hub	1	M.S.

Steering Column	1	AISI 4130
Steering Column Hub	1	M.S.
Bearing	2	SS
Pitman arm	1	M.S.
Tie Rods	2	AISI 4130
Spherical Rod Ends	4	SS
C-Clamps	2	M.S.
Stub Axle(Upright)	2	M.S.

Table 2.1

Working principle:

The steering system of our go-kart operates using a basic four-bar linkage mechanism, which is commonly used in karting due to its simplicity, low weight, and effective control.

When the driver rotates the steering wheel, the motion is transferred to the steering column. The lower end of the steering column is connected to a pitman arm or steering arm, which converts the rotary motion of the column into lateral (side to side) motion.

This lateral movement is passed to the tie rods, which are connected to both sides of the pitman arm and extend outward to the steering knuckles or uprights mounted on the front wheels. As the tie rods push or pull, they cause the wheels to pivot about their respective kingpins.

The pivoting of the front wheels allows the go-kart to turn left or right. The angles at which the wheels turn are carefully designed to follow the Ackermann steering geometry, which helps reduce tire slip and improves

turning efficiency by ensuring both wheels follow their correct turning paths during a corner.

To ensure safety and rule compliance:

- The system is limited to 7° of free play at the steering wheel.
- Mechanical stops prevent over-rotation that could invert the linkage.
- All components are made from metal as required by the rulebook.
- The turning radius of the kart remains under the 3-meter maximum limit as defined by IKR.

Include the pictures of design, component, and assembly. (If any):

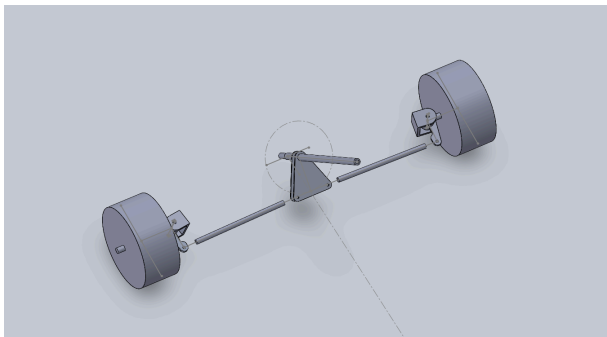


Figure 2.1 Steering Assembly

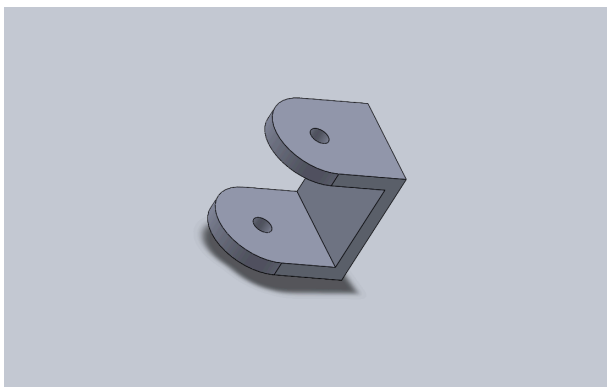


Figure 2.2 C-clamp

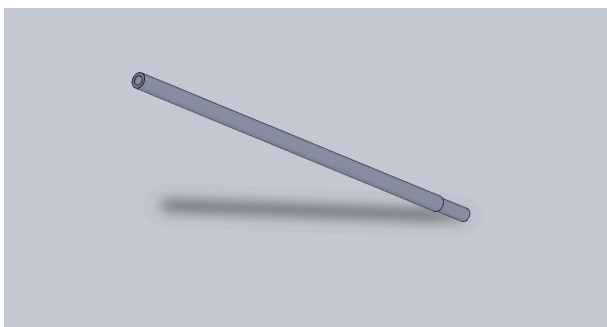


Figure 2.3 Steering Column

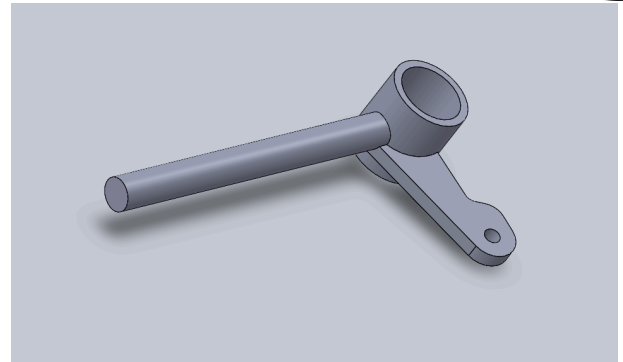


Figure 2.4 Upright



Figure 2.5 Pitman Arm

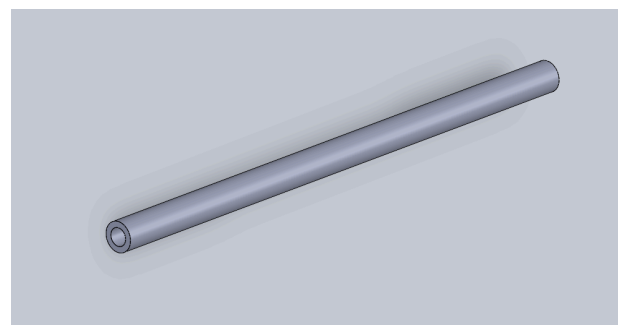


Figure 2.6 Tie Rod

Formulation and Calculations:

Front track $T = 1050 \text{ mm}$

Wheel-base $W = 1299 \text{ mm}$

Design turning radius $R = 1774.75 \text{ mm}$ (1.775 m)

Front axle load = 70 kg $\Rightarrow$ 343 N per wheel

Assumed CG height = 250 mm

Cornering speed = 5 m s^{-1} ($\approx 18 \text{ km h}^{-1}$)

Caster = 8°

King-Pin Inclination (KPI) = 8°

Ackerman Angle(α) = $\tan^{-1}(0.5 \cdot T/W) = 22.0^\circ$

1. Turning Radius and Steer Angles:

Outer Angle:



$$\theta_0 = \tan^{-1}(W / R + T/2) = 29.46^\circ$$

Outer Turning Radius:

$$R_0 = W / \sin(\theta_0) = 2641 \text{ mm}$$

Inner angle:

$$\theta_i = \tan^{-1}(W / R - T/2) = 46.11^\circ$$

Inner Turning Radius:

$$R_i = W / \sin(\theta_i) = 1803 \text{ mm}$$

Actual turning radius:

$$R = (T/2) + (W / \sin(y))$$

Where

$$y = (\theta_0 + \theta_i) / 2$$

$$R = 2645 \text{ mm}$$

Ackerman Percentage:

$$\text{Ackerman Percentage} = (\theta_i / \alpha) * 100 = 209.590$$

2. Steering Efforts:

Lateral Acceleration:

$$AL = v^2 / R = 14.09 \text{ m s}^{-2} (1.44 \text{ g})$$

Weight transfer during cornering:

$$\Delta w = \text{Front Load} \times AL \times \text{Height of C.G.} / \text{Front Track Width} = 235 \text{ N}$$

Weight on inner wheel (W_{in}):

$$W_{in} = \text{Load on Front wheel} - \Delta w$$

$$= 109 \text{ N}$$

Weight on outer wheel (W_{out}):

$$W_{out} = \text{Load on Front wheel} + \Delta w$$

$$= 578 \text{ N}$$

Lateral Force on inner wheel (L_i):

$$= \text{Weight on inner wheel} \times AL$$

$$= 156 \text{ N}$$

Lateral Force on outer wheel (L_o):

$$= \text{Weight on outer wheel} \times AL$$

$$= 830 \text{ N}$$

Moment due to lateral force on wheels:

$$M_{lateral} = \text{Lateral Force} \times \text{wheel radius} \times \tan(\text{caster angle})$$

$$M_{lateral I} = 2783 \text{ Nmm}$$

$$M_{lateral O} = 14817 \text{ Nmm}$$

Moment at Kingpin due to Lateral Force:

$$M_{lateral k} = M_{lateral I} + M_{lateral O} = 17600 \text{ Nmm}$$

Moment due to self-aligning torque on wheels:

$$M_{SAT} = \text{Lateral Force} \times \text{Contact Patch} / KPI$$

$$M_{SAT I} = 1754 \text{ Nmm}$$

$$M_{SAT O} = 9339 \text{ Nmm}$$

Moment at Kingpin due to Self-aligning torque:

$$M_{SAT k} = M_{SAT I} + M_{SAT O} = 11093 \text{ Nmm}$$

Net Moment at Kingpin:

$$M_{net} = M_{lateral k} + M_{SAT k}$$

$$= 28693 \text{ Nmm}$$

Force in Pitman Arm:

$$= \text{Net moment at Kingpin} / \text{Length of Steering arm}$$

$$= 98.9 \text{ N}$$

Horizontal force along Tie Rods:

$$= \text{Force in steering arm} / \cos(\text{Ackerman Angle}) = 106.7 \text{ N}$$

Moment at Steering column:

$$= \text{Horizontal Force along tie rod} \times \text{Tie rod radius}$$

$$= 10465 \text{ Nmm}$$

Steering Effort:

$$= \text{Moment at column} / \text{Steering wheel radius}$$

$$= 82.4 \text{ N}$$



Final Steering Effort (one handed effort):

$$= \text{Steering Effort} / 2$$

$$= 41.2 \text{ N}$$

Results. (Turning radius, steering ratio, steering angle, driver effort etc.):

Wheel base is 1299 mm

Front track width 1050 mm

Ackerman angle 22.0°

Outer angle is 29.46°

Outer turning radius is 2641 mm

Inner turning angle is 46.11°

Inner turning radius 1803 mm

Actual turning radius 2645 mm

Steering effort : 82.4 N

Final steering effort (single handed effort) : 41.2 N

TRANSMISSION DESIGN REPORT

Introduction

The Transmission System is responsible for delivering motion and power from the motor to the rear shaft. In our case, the transmission system that has been used is the chain drive. The objective of this section of the report is two-fold. Firstly, selection of an appropriate chain and sprocket pair, and secondly, determining a safe, threshold diameter for the shaft on a material strength basis.

Rulebook Constraints

- The transmission to be used in the Go-Karts must be rear wheel drive only.
- The teams are free to use any type of transmission system – gearbox/ CVT/ Chain and sprocket attached to a centrifugal clutch.
- The teams are free to use any sort of designs i.e. the use of differential, through the axle, the wheel mounting hub or by any other means.
- There is no limitation with the sprocket ratio installed in the kart. Teams are

allowed to make changes to the sprocket ratio as per the requirements.

Force Consideration

1. Chain and Sprocket

Tension force in the chain will act on the rear shaft.

2. Normal Ground Reaction

This force will act vertically upward on the rear wheels. For simplicity of calculation, we assume 75-80% of the total vehicle weight to be acting on the rear portion of the kart, this is borne equally by both the rear wheels.

3. Twisting Moment/ Torque

The twisting moment is a consequence of the torque being transmitted from the motor to the rear shaft (driven shaft).

4. Bearing Reactions

The forces sustained on the bearings are calculated by applying the Conditions of Equilibrium (Moment balance and force balance), given that the calculations are done in a static scenario.

Working Principle

The motor delivers a power of 4kW at 3000rpm. The rotational velocity gets stepped down as per the velocity ratio (shown ahead). For argument sake, the friction losses in transmitting the power are neglected.

Design Consideration

Preliminary Assumptions: In order to simplify the calculations, we shall neglect the self weight of members such as the brake disc and the driven sprocket, and also that of the shaft.

The forces and torques that were previously stated will impose a state of combined bending and twisting of the rear shaft. Therefore, in the calculation of a suitable threshold diameter for the shaft, we shall consider both these effects.

In order to fetch appropriate values for the stresses that will ensure a modicum of safety in the design, we have invoked the ASME Code for Shaft design.

$$\tau_{\max.} = 0.30 S_{yt}$$

$$\tau_{\max.} = 0.18 S_{ut} \text{ (whichever is minimum)}$$

Furthermore, we reduce 25% of the given value to compensate for loss of material due to cutting of keyways.

Material Consideration: For the purpose of calculation, we have considered the material to be SAE 4130 Chromoly steel which has $S_{ut} = 540\text{MPa}$ and $S_{yt} = 460\text{MPa}$.

The Theory of Failure used herein, is the Maximum Shear Stress Theory (Tresca, Coulomb, Guest), which is suitable for ductile materials.

Component Selection

For power transmission, the chain selected is ISO 08B.

Static Structural Analysis in Ansys

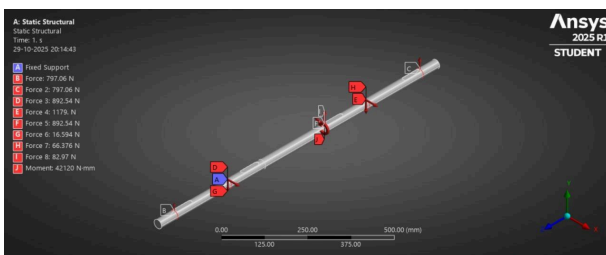


Figure 3.1. Forces, moments, and supports applied to the shaft

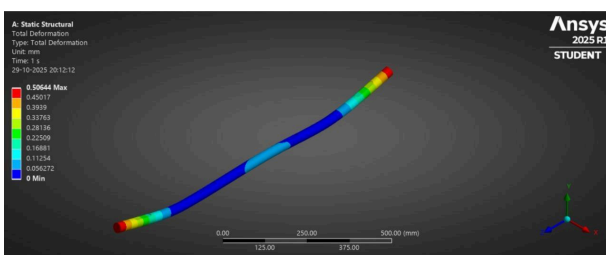


Figure 3.2. Total Deformation (exaggerated scale)

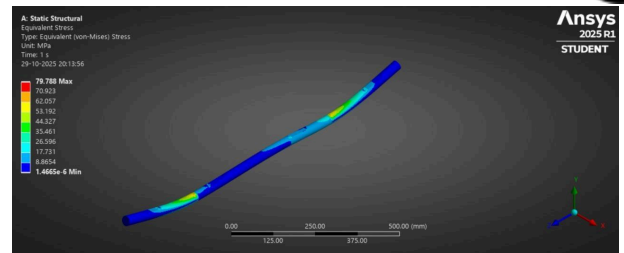


Figure 3.3. Equivalent Stress

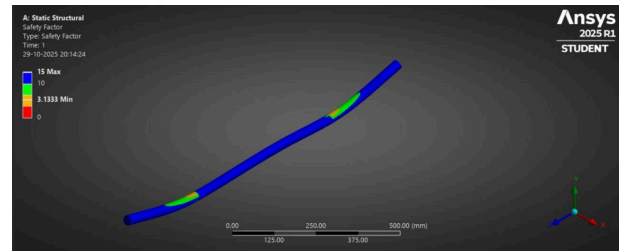


Figure 3.4. Safety Factor

Formulations and Calculations

1. Chain and Sprocket

Geometric Terminologies:

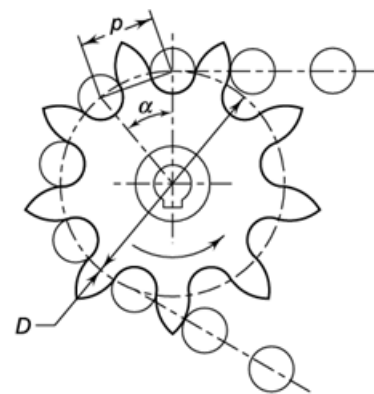


Figure 3.5. Chain Sprocket

D : Pitch circle diameter of the sprocket. The pitch circle diameter of the sprocket is defined as the diameter of an imaginary circle that passes through the centres of link pins as the chain is wrapped on the sprocket.



p : Pitch

α : Pitch angle.

$$\alpha = \frac{360}{z}$$

z: Number of teeth on the sprocket.

From the figure, it can be proved that

$$\sin\left(\frac{\alpha}{2}\right) = \frac{(p/2)}{(D/2)} \quad \text{or} \quad D = \frac{p}{\sin\left(\frac{\alpha}{2}\right)}$$

The velocity ratio i of the chain drives is given by,

$$i = \frac{n_1}{n_2} = \frac{z_2}{z_1}$$

n_1, n_2 = speeds of rotation of driving and driven shafts (rpm)

z_1, z_2 = number of teeth on driving and driven sprockets.

The average velocity of the chain is given by,

$$v = \frac{\pi D n}{60 \times 10^3} \quad \text{or} \quad v = \frac{z p n}{60 \times 10^3}$$

The length of the chain is always expressed in terms of the number of links, or

$$L = L_n \times p$$

where

L = length of the chain (mm)

L_n = number of links in the chain

The number of links in the chain is determined by the following approximate relationships:

$$L_n = 2\left(\frac{a}{p}\right) + \left(\frac{z_1 + z_2}{2}\right) + \left(\frac{z_2 - z_1}{2\pi}\right)^2 \times \left(\frac{p}{a}\right)$$

where,

a = centre distance between axes of driving and driven sprockets (mm)

It is always preferred to have an 'even' number of links, since the chain consists of alternate pairs of inner and outer link plates. On the other hand, the number of teeth on the sprocket is typically an odd number.

In order to approximate the dimensions of the sprocket, we use the above, and other empirical formulae:

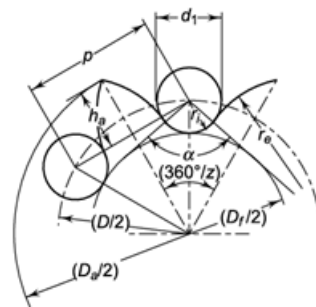


Figure 3.6

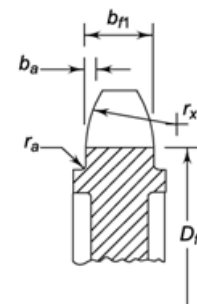


Figure 3.7

Standardisation of the Pitch: For our case, we use a sprocket/chain with a pitch of 12.7mm (ISO chain 08B)

Some of the standard dimensions corresponding to the pitch value:

Roller diameter (d_1) = 8.51mm

Width (b_1) = 7.75mm



Determining the Pitch Circle Diameters:

a. Driving Sprocket

$$z_1 = 17 \text{ teeth}$$

$$D_1 = p/\sin(180/z_1) = 69.11\text{mm}$$

$$\approx 70\text{mm}$$

b. Driven Sprocket

$$z_2 = 43 \text{ teeth}$$

$$D_2 = p/\sin(180/z_2) = 173.98\text{mm}$$

$$\approx 174\text{mm}$$

Velocity Ratio

The velocity ratio of the driven sprocket to the driving sprocket, i , is given as:

$$i = z_1/z_2 = n_2/n_1 = 2.53$$

$$\text{Given that } n_1 = 3000\text{rpm}$$

$$n_2 = 1186.05\text{rpm}$$

2. Shaft Calculations

Transmission shafts are subjected to axial tensile force, bending moment or torsional moment or their combinations. Most of the transmission shafts are subjected to combined bending and torsional moments. The design of transmission shafts consists of determining the correct shaft diameter from strength and rigidity considerations.

When the shaft is subjected to pure bending moment, the bending stresses are given by,

$$\sigma_b = \frac{M_b y}{I} = \frac{M_b \left(\frac{d}{2} \right)}{\left(\frac{\pi d^4}{64} \right)} \quad \text{or}$$

$$\sigma_b = \frac{32 M_b}{\pi d^3}$$

When the shaft is subjected to pure torsional moment, the torsional shear stress is given by,

$$\tau = \frac{M_t r}{J} = \frac{M_t \left(\frac{d}{2} \right)}{\left(\frac{\pi d^4}{32} \right)} \quad \text{or} \quad \tau = \frac{16 M_t}{\pi d^3}$$

Shafts are made of ductile material like steel and therefore, the appropriate theory for failure is the Maximum Shear Stress Theory. In order to apply said theory, we calculate the Equivalent Torsional Moment and essentially use the formula for pure torsion.

$$\tau_{\max.} = \sqrt{\left(\frac{16 M_b}{\pi d^3} \right)^2 + \left(\frac{16 M_t}{\pi d^3} \right)^2}$$

$$\tau_{\max.} = \frac{16}{\pi d^3} \sqrt{(M_b)^2 + (M_t)^2}$$

Where, the expression

$\sqrt{(M_b)^2 + (M_t)^2}$ is the “Equivalent Torsional Moment”

a. Torsional Moment

Owing to the velocity ratio, the velocity of the driving sprocket and therefore the shaft is 1186.05rpm. Assuming no power loss, the power delivered to the shaft from the motor is 4kW.

$$\text{Since, } M_t = 60(\text{Power})/2\pi n_2$$

$$M_t = 32.2 \text{ Nm} = 32200 \text{ Nmm}$$

b. Bending Moment

Shaft Schematic:

The following is a schematic of the shaft with mountings that impose transverse forces on the shaft.

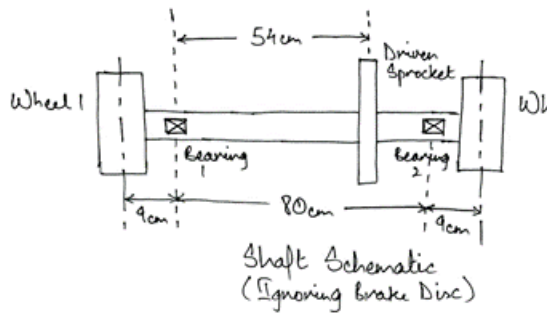


Figure 3.8: Shaft Schematic

Free Body Diagrams and Bending Moment Diagrams

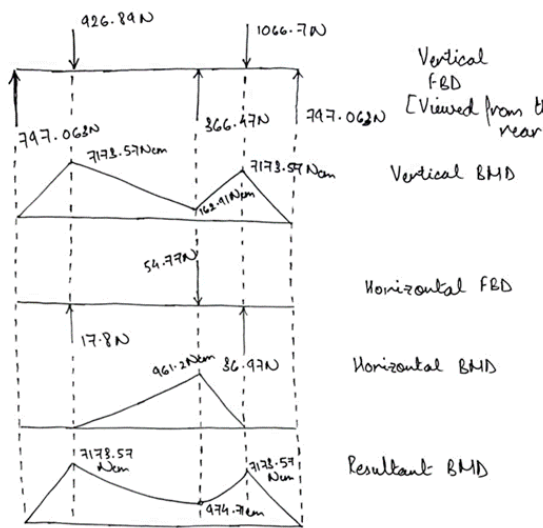


Figure 3.9: SFD and BMD of Shaft

In order to calculate the above, the chain tension was separately calculated and resolved into two components, yielding a horizontal and vertical Free Body Diagram.

The maximum value of bending moment is used to proceed with the calculation, i.e. $M_b = 7173.57 \text{ Ncm}$

Recalling the ASME convention, the permissible value of shear stress can be found as:

$$\tau = \min(0.3S_{yt}, 0.18S_{ut}) = \min(138, 97.2)$$

Compensating for the cutting of keyways

$$\tau_{\max} = 0.75\tau = 72.9 \text{ MPa}$$

Using the formula:

$$\tau_{\max} = \frac{16}{\pi d^3} \sqrt{(M_b)^2 + (M_t)^2}$$

$$d = 17.64 \text{ mm}$$

$$d_{\min} = 18 \text{ mm (say)}$$

Results

For the given material, and given force considerations, a safe, minimum threshold diameter is 18mm. In practice, of course the shaft diameter will be larger than this value. The calculation does however assure a modicum of safety in the design.

BRAKING SYSTEM DESIGN REPORT

Rulebook Constraints:

C.3.7.1 Braking System

C.3.7.1.1 Teams must use the hydraulic braking system. "Brake-by-wire" systems are prohibited.

C.3.7.1.2 The brakes must be able to act on both the rear wheels.

C.3.7.1.3 The hydraulic circuit must have its own fluid reservoir, either by use of separate reservoirs or by the use of a dammed, OEM-style reservoir.

C.3.7.1.4 Team must use steel braided brake hose and brake lines.

C.3.7.2 Brake Pedals

C.3.7.2.1 The brake pedal must be fabricated from steel or Aluminium or machined from steel, Aluminium or Titanium.

C.3.7.2.2 Pedal should only be operated from driver's foot and use of hand operated levers for braking mechanism is not allowed.

C.3.7.2.3 Teams need to submit the report of calculation, Selection of Material, and safety of brake pedal.

C.3.7.2.4 There should not be free play at pedals.



C.3.7.2.5 Brake pedal should be able to withstand a force of 2000 N.

C.3.7.3 Pedal Locks

C.3.7.3.1 The pedal travel should be restricted to some distance by some kind of locking mechanisms.

C.3.7.4 Brake over travel Switch

C.3.7.4.1 A brake pedal over-travel switch must be installed on the car as part of the shutdown system and wired in series with the shutdown buttons. This switch must be installed so that in the event of brake system failure such that the brake pedals over travels, it will result in the shutdown of the system, which will eventually help controlling the system.

C.3.7.5 Brake Light

C.3.7.5.1 The kart must be equipped with a red colour brake light.

C.3.7.5.2 The brake light itself must be rectangular, triangular or near-round shape with a minimum shining surface of at least 15 cm sq.

C.3.7.5.3 Each brake light must be clearly visible from the rear in very bright sunlight. It must be on the top-center of Roll hoop

Force considerations

We have considered a total clamping force of 2500N based on the dimensions and specifications of our calliper, given below are the specifications of the same:

- Number of Pistons: 1
- Diameter of Each Piston: 30mm

The friction force was calculated by multiplying our assumed friction factor (0.4) with the clamping force which came out to be 2,000N

Working principle

Disc Brakes convert kinetic energy to slow down or stop a vehicle, When the brake pedal is pressed, hydraulic pressure forces brake pads against a rotating disc (rotor), creating friction that slows the wheel. This friction generates heat, which is then dissipated into the surrounding air.

Following are a few important points concerned with the working principle of disc brakes:

1. Hydraulic Pressure:

- When the brake pedal is pressed, it activates the master cylinder, which sends pressurized brake fluid through brake lines to the calipers at each wheel.

2. Caliper Action:

- The hydraulic pressure pushes pistons within the calipers, which in turn force the brake pads against the rotating disc.

3. Friction and Heat:

- The brake pads, made of friction material, grip the disc, causing friction. This friction converts the kinetic energy of the rotating wheel into thermal energy (heat), slowing down the wheel and the vehicle.

4. Release:

- When the brake pedal is released, the pistons retract, and the brake pads move away from the disc, allowing the wheel to rotate freely.

Design Considerations:

- We designed the brake disc taking into consideration the minimum compatible diameter of the brake calliper we had chosen, the brake calliper is designed for a 230 mm diameter disc but due to space constraints we had to design the disc to an approximate 180 mm diameter which seemed to fit well with the calliper.

Component Selection:

Brake Calliper : KBX Single Piston

Master Cylinder: KBX

Brake Disc : Own Design



Figure 4.1. Brake Disc



Figure 4.2 KBX Single Piston Caliper

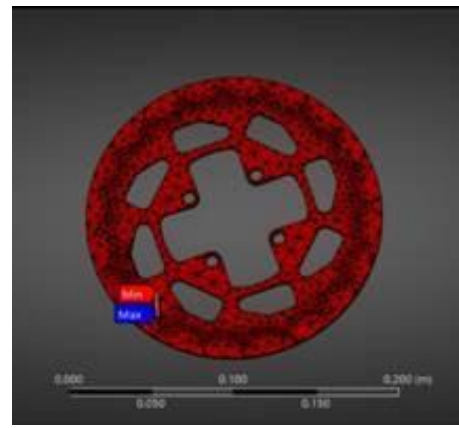


Figure 4.5 Safety Factor Analysis (FOS=15)

Part analysis (Brake Disc)

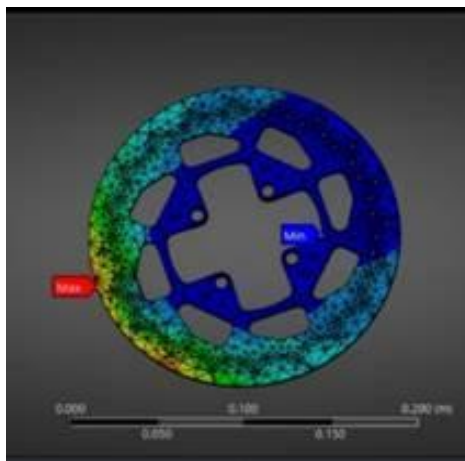


Figure 4.3. Total Deformation Analysis

Part Analysis (Brake Pedal):

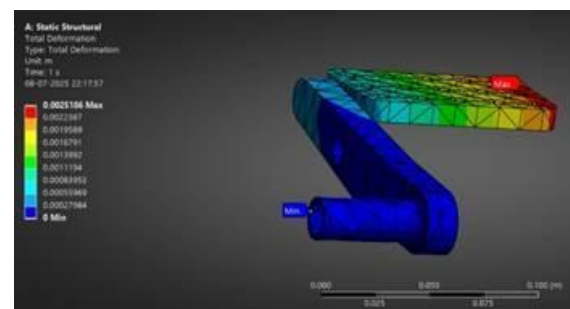


Figure 4.6. Total Deformation Analysis

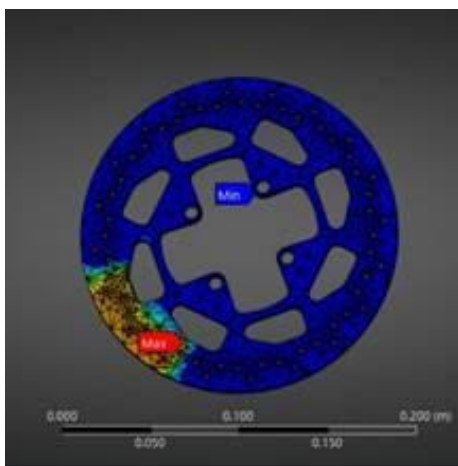


Figure 4.4. Stress Analysis

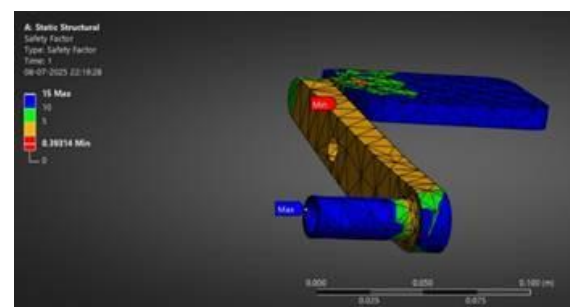


Figure 4.7. Safety Factor Analysis (FOS = 15)

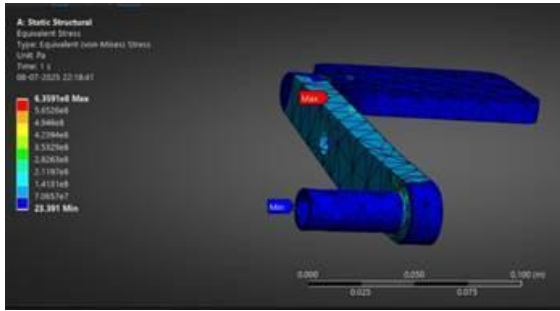
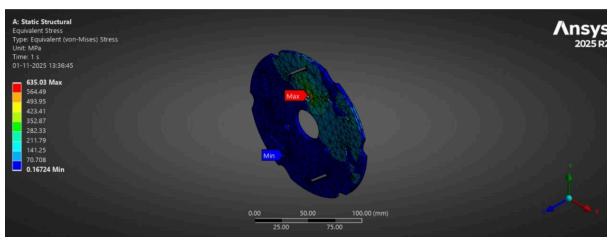
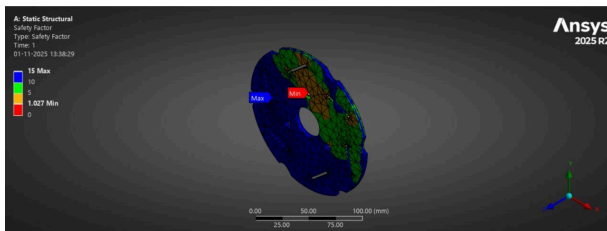
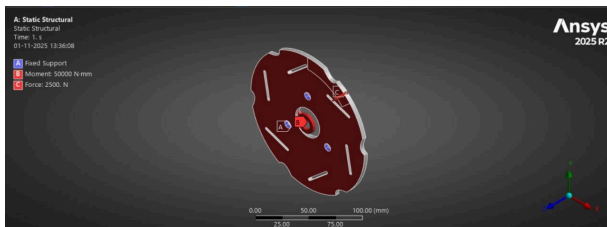
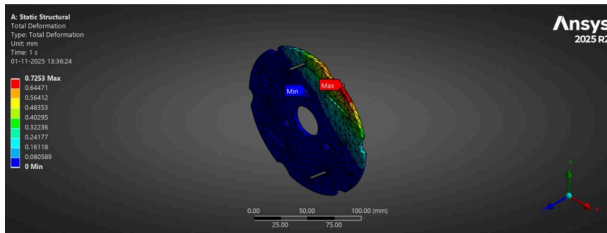


Figure 4.8. Stress Analysis



Formulation and Calculations:

Area of each braking piston:

$$A = \pi \times (12.5 \text{ mm})^2 \approx 490 \text{ mm}^2$$

$$10,000,000 \text{ N/m}^2 \times 0.00098 \text{ m}^2 = 9,800 \text{ N}$$

For aggressive braking let's assume it to be 10,000N

Let:

$$F_{\text{Clamp}} = F_{\text{piston, left}} + F_{\text{piston, right}}$$

$$F_{\text{Clamp}} = 5,000 \text{ N}$$

$$\mu (\text{Co-efficient of Friction}) = 0.4$$

Calculate the Braking Force:

$$F_{\text{braking}} = \mu \times F_{\text{clamp}}$$

$$F_{\text{braking}} = 0.4 \times 5,000 \text{ N} = 2,000 \text{ N}$$

Results:

Stopping Distance = 3.91m from 45 km/h to 0 km/h

Stopping Time = 0.625s

MOTOR DESIGN REPORT

- Introduction
The motor is a principal part of the tractive system of the electric kart. It converts the electrical energy supplied by the accumulator



2. Gear ratio: 37/11
3. Wheel torque = $12.8 \times (37/11) = 43.05$ Nm

Speed:

1. Motor RPM: 3000
2. Wheel RPM = $3000 \times (11/37) = 891.891$ RPM
3. Wheel radius = 0.1397 m

Therefore, Linear speed = $750 \times 2\pi \times 0.1397 / 60 \approx 10.972$ m/s ≈ 39.499 km/h

Power:

1. $P = V \times I \times \eta = 60 \text{ V} \times 66.66 \text{ A} \times 0.85 \approx 3.9 \text{ kW}$

- Tabulations.

S.No.	Feature	Dimension (mm)
1	Length (L)	302.8
2	Shaft Length	53.15
3	Shaft Diameter	24
4	Key Length	40
5.	Key Width	8
6.	Key Height with shaft	27

Table 5.1: Motor Dimensions

Parameter	Value
Type	BLDC
Model	6D-4
Rated Power	4 kW
Voltage	60V
Rated Current	86.66 A
Rated Torque	12.8 Nm
Rated RPM	3000 RPM
Peak Current	134.4 A
Cooling	Air
Efficiency	~85%

Table 5.2: Motor Specifications

- Results. (Power, Voltage, Current, Torque, RPM, Efficiency etc.)

1. Verified compliance with IKR rulebook.
2. Performance at 60 V: 12.8 Nm, 4.25 kW output, 3000 RPM, 66.66 A.
3. Capable of 40 km/h speeds with 37:11 drive ratio
4. Matches torque and acceleration goals for kart

MOTOR CONTROLLER DESIGN REPORT

- Introduction

This section details the selection and application of the motor controller for the electric race kart developed for the Indian Karting Race 2025. It includes considerations for rulebook compliance, load requirements, operating principle, design integration, and electrical behavior.

- Rulebook Constraints

C.4.11.1 The tractive system motor must be connected to the accumulator through a motor controller.

C.4.11.2 Controller must provide over & low voltage protection, over temperature protection, over current protection & brake protection to motor.

C.4.11.3 Bypassing the control system and connecting the tractive system accumulator directly to the motor is prohibited.

C.4.11.4 The controller voltage should match the voltage of the battery and motor. The controller current rating should be lower or equal to the continuous current output of the battery. The system should be designed in such a way that all the components are properly tuned with each other.

- Load consideration.

1. Battery Output: 60 V, 100 A max continuous
2. Motor Specs: 4 kW, 66.66 A rated, 3000 RPM
3. Controller must handle full-load motor current with thermal headroom
4. Ambient operation: 25–40°C (air-cooled)

- Working principle



The Kelly KLS7230N is a sinusoidal, programmable, brushless DC motor controller. It uses:

1. FOC (Field-Oriented Control) for smooth torque delivery
2. Hall sensor feedback to detect rotor position
3. PWM output to 3-phase motor windings
4. Analog input (0.8–4.2 V) from throttle sensor
5. ENABLE line for safety interlock (must be HIGH to run)

• Design Considerations

1. Voltage Rating: ≥ 72 V (safely supports 60 V input)
2. Current Rating: Configured to 100 A continuous
3. Compatible with Hall-based BLDC motor
4. ENABLE input controlled via brake-overtravel switch
5. Pre-charge and contactor control in circuit design
6. Communication via UART (for config & diagnostics)

• Component Selection.

1. Controller Model: Kelly KLS7230N
2. Supported Motor Types: BLDC, PMSM, ACIM
3. Operating Voltage Range: 24–72 V
4. Max Continuous Current (configurable): 150 A
5. Max Phase Current: ~ 300 A
6. Logic Input: 5 V/12 V ENABLE, Analog Throttle
7. Cooling: Aluminum heatsink with air-cooling.

• Formulation and Calculations.

Power handled:

1. Input: $P = V \times I = 60 \text{ V} \times 100 \text{ A} = 6 \text{ kW}$ (max input capability)
2. Controller efficiency (typical): $\sim 95\%$
3. Output Power: $\approx 5.7 \text{ kW}$

• Tabulations.

Parameter	Value
Model	Kelly KLS7230N
Input Voltage	60 V (used), rated for 72 V
Continuous Current	Configured to 100 A
Max Phase Current	~ 300 A
Throttle Input	0.8–4.2 V Analog
ENABLE Logic	Active HIGH
Sensor Feedback	Hall (UVW)
Communication	UART
Cooling	Air-cooled, aluminum

Table 6.1: Motor Controller Specifications

• Results.

1. Confirmed controller compatibility with 6D-4 motor
2. Safe current levels within battery spec (≤ 100 A continuous)
3. Smooth throttle response using Hall sensor input
4. Reliable ENABLE signal via low-voltage shutdown chain

BATTERY DESIGN REPORT

• Introduction

This section presents the design, configuration, and analysis of the battery pack used in the electric race kart for Indian Karting Race (IKR) 2025. It includes performance estimates, safety provisions, and calculations to ensure compatibility with all powertrain components.

• Rulebook Constraints

C.4.12.1.1 Teams are allowed to use pre-manufactured battery pack available in the market, if self-fabricated or assembled battery pack is intended to use by team, then team needs to provide the IP 67 certificate and battery health certificate at the time of Technical Inspection issued from authorized certifying agency.

C.4.12.1.2 Battery pack should meet all weather conditions to ensure that it should not fail.

C.4.12.1.3 The teams can use either of the following types of the batteries – SMF, Li-ion.



C.4.12.1.4 Battery pack should be IEC 60529 IP67 certified.

C.4.12.1.5 If the team is using an old battery pack, you need to submit the Battery Health Certificate.

C.4.12.1.6 Teams must use a battery of the following configuration:

Maximum Rated voltage: 144V

Maximum Capacity: 150AH

- Load consideration.
 1. Motor rated current: 83.33 A
 2. Controller configured current: 100 A
 3. Vehicle full-load demand ≈ 4.5 kW
 4. Peak current may spike during acceleration, therefore the BMS and cell ratings are selected accordingly.
- Working principle

The battery pack consists of multiple 18650 or 21700 Li-ion cells arranged in a series-parallel structure:

 1. HLY 26650-5000mAh Series cells determine pack voltage .
 2. Parallel cells determine capacity and current rating
 3. BMS monitors cell voltage, temperature, and current
 4. Power flows to controller through safety chain (fuse, contactor, pre-charge)
- Design Considerations
 1. Required voltage: ~ 60 V nominal, 67.2 V max
 2. Required current: ≥ 100 A continuous
 3. Capacity target: 90 Ah
 4. Cell configuration: 17S18P
 5. Cell protection: BMS with balancing and cutoff
 6. Thermistor integration for temperature monitoring
- Component Selection.
 1. Model: EXCEL 1706090
 2. Voltage: 60V
 3. Max. Continuous Current: 100A
 4. Max. Peak Current: 150A
 5. Capacity 150Ah
 6. Equipped with Short Circuit Protection, Voltage Cutoff, High Temperature Sense Cut-Off, Visual

Heat Indicator, Passive Cell Balancing.

- Formulation and Calculations.
 1. Total Energy:

$$E = V \times Ah = 60 \text{ V} \times 90 \text{ Ah} = 5.4 \text{ kWh}$$
 2. Battery Runtime (at avg 4 kW load):

$$\text{Runtime} = E / P = 5.4 \text{ kWh} / 4 \text{ kW} = 1.35 \text{ hours}$$
 3. Voltage per cell:

$$3.7 \text{ V nominal} \times 17 = 62.9 \text{ V}$$

$$\text{Max charge voltage: } 4.25 \text{ V} \times 17 = 72.25 \text{ V}$$
 4. Parallel Discharge Current Capability:

$$18P \times 5 \text{ A} = 90 \text{ A}$$

- Tabulations.

Parameter	Values
Chemistry	Li-ion
Configuration	17S18P
Voltage (Nominal)	60 V
Capacity	90 Ah
Energy	5.4 kWh
Continuous Current	100 A
Peak Current	150 A
Cooling	Passive, air vented

Table 7.1: Accumulator Specifications

- Results.(Power, Voltage, Current, Charging rate, Efficiency etc.)
 1. Battery meets power and energy requirements for a healthy duration.
 2. Peak current capability is compliant with motor/controller needs



3. Integrated BMS ensures protection, balancing, and thermal cutoff
4. Power: 5.4 kWh
5. Continuous Current: 100A
6. Voltage: 60V

WIRING DIAGRAM AND ACCESSORIES DESIGN REPORT

- Introduction

This report provides a complete overview of the wiring system and key electrical accessories used in the electric race kart designed for the Indian Karting Race (IKR) 2025. It details the power and signal distribution layout, shutdown logic, and integration of supporting systems including relays, switches, and sensors.

- Rulebook Constraints

C.4.12.6.6 The battery pack must have a proper short circuit protection. Teams should use good quality working Mini circuit breakers (MCB) for protecting it from high loads.

C.4.12.6.7 The current rating of MCB should not be greater than the cable current carrying capacity. It should be greater than or equal to the maximum current that the system is carrying. Do not use a MCB lower than the total current of the circuit.

C.4.12.6.8 Team should choose MCB as per the vehicle requirement and must understand the MCB ratings.

C.4.15.1.1 There should be three kill switches in the Go-Kart.

C.4.15.1.2 They should be placed in such a way that one can be easily accessed by the driver and the other two should be mounted on either side of the roll hoop.

C.4.15.1.3 Kill switch must be of bright red color so that it can be clearly visible from a long distance.

C.4.15.1.4 It should be a push to off type, it should not be a simple push button which retraces after the button is pressed.

C.4.15.1.5 The kill switch used should be as shown in the figure below.

C.4.15.1.6 Home appliance switches should not be used as kill switches.

C.4.15.4.1 All electrical systems must be appropriately fused. Batteries should be shielded properly from the driver.

C.4.15.4.2 Battery terminal should be insulated using proper battery plastic caps or by gel, use

of tapes, cardboard or plastic sheet, rubber sheet or insulation tape is not allowed.

C.4.15.4.3 Since the battery has very low internal impedance, instantaneous high currents can flow which can seriously damage the battery.

C.4.15.4.4 Teams should mandatorily use an MCB (mini circuit breaker) which should be instantaneously able to isolate the battery from the other electronic devices to protect the battery from an overload.

C.4.15.4.5 The current rating of the fuse should be greater (15% to 20%) than the current carrying through the wire on which it is installed.

C.4.15.4.6 The fuse holder must be rigidly mounted on the chassis.

C.4.15.4.7 The tractive system must be appropriately fused.

C.4.12.1.6 Teams must use a battery of the following configuration:

Maximum Rated voltage: 144V

Maximum Capacity: 150AH

- Design Considerations

1. Safe and modular layout of the high-voltage and low-voltage sections.
2. Efficient routing of power, logic, and control signals.
3. Clear grounding system separation between TS and GLV.
4. Use of high-current relays, appropriately rated fuses, and emergency switches.
5. Reliable triggering of contactors and shutdown using kill switches..
6. Interlock logic using ENABLE pin of controller
7. Regulated +12 V supply for throttle, relays, and logic.

- Circuit design

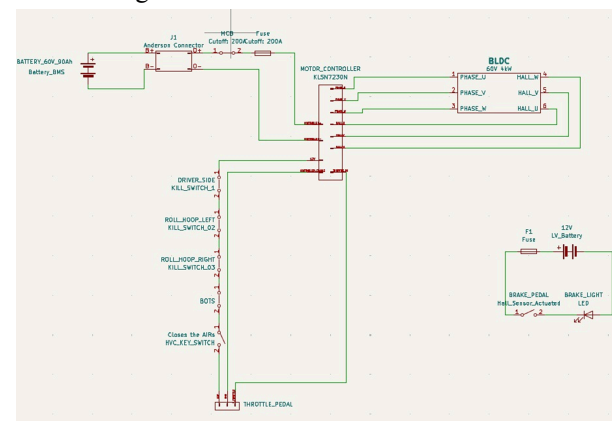


Figure 8.1: Circuit Schematic

- Formulation and Calculations.

1. Pre-charge Resistor: 220 Ω , 10 W $\rightarrow$ limits inrush current to <1 A
2. 180 A fuse and MCB is used to protect the circuit and its components in the event of a current surge over the 150A limit.

- Tabulations.

Component	Specification
Pre-charge Resistor	220 Ω , 10 W
Main Contactor	60V
K1 Pre-charge Relay	SPST, 12 V coil
Kill Switches	3 $\times$ NC switches in series
Brake Overtravel Switch	NC switch
DC-DC Converter	60V to 12V, 5 A
Fuse (F1)	180 A

Table 8.1: Circuit Component Specifications.

- Results.

1. Complete and rulebook-compliant shutdown system implemented.
2. Isolation between TS and GLV is achieved via DC-DC converter.
3. Controller reliably disables when Brake Overtravel Switch/Kill Switch is opened.
4. All devices powered safely through +12 V GLV rail with no overload.
5. Diagrams and component placement ensure easy troubleshooting.

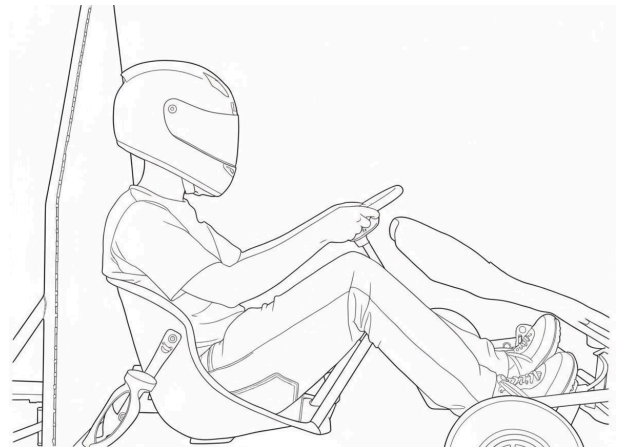
Component Packaging



The rear part of the kart is the power producing center of the kart. Best techniques were adopted to package the powertrain in a compact form while adhering to IKR regulations. The battery has been positioned in-between

the motor and the firewall, this proves to be the most optimal solution as the battery is positioned vertically and the weight distribution of the kart is balanced.

Occupant Packaging



REFERENCES

- [1] Edward James Goodman - "Race Car Dynamics & Design Applied to Formula Student"
- [2] V B Bhandari "Design of Machine Elements Third Edition", McGraw Hill, 2010, Chapter 14, pp. 544-553.

